Do Pruning and Quantization Compose? Order and Interaction in Post-Training Compression of Language Models

[Authors and affiliations to be confirmed]

Abstract. Post-training pruning and quantization are routinely stacked to compress language models, under the implicit assumption that their errors are independent. We test this assumption inside one optimal-brain-surgeon framework, comparing joint, prune-then-quantize and quantize-then-prune compression at equal storage cost on three models (135M–0.5B), and we re-analyse published 7–8B results. A decomposition of the layer error predicts sub-additivity whenever the pruning mask is chosen before the quantization grid. Experiments confirm it in all 36 mask-first cases. Quantizing first is super-additive at 4 bits and gives the worst perplexity in 16 of 18 settings, up to twice the best one.

1 Introduction

Pruning removes weights; quantization shortens the remaining ones. Both have accurate one-shot, post-training versions for large language models (LLMs) built on the same second-order principle: SparseGPT [1] for pruning and GPTQ [2] for quantization, both descendants of Optimal Brain Surgeon (OBS) [3, 4]. Stacking them is attractive because the gains multiply [5]. In earlier work [6] we showed that, at an equal theoretical compression rate (TCr), 25% pruning with 4-bit quantization retains about 20% more performance than pure 3-bit quantization on 7–8B models. That study, like most practice, applied the two methods one after the other and assumed that their errors do not interact.

This assumption is not innocuous. Harma et al. [7] proved that sparsity and max-scaled quantization are non-orthogonal and that sparsity should come first. Joint methods [8, 9] report that naive stacking leaves accuracy on the table. What is missing is a direct measurement, inside the Hessian-based post-training pipeline used in practice, of *how* the two errors combine and of whether the answer depends on model size.

We answer three questions. **Q1:** is the combined layer error the sum of the individual errors? **Q2:** at fixed TCr, does the order of the operations matter? **Q3:** does the interaction change with scale? Our contributions are: (i) a single OBS sweep that implements pruning, quantization and their three combinations with identical machinery, so that only the order differs; (ii) a decomposition of the combined layer error that explains when it is sub-additive; (iii) a controlled study of 93 runs on three models (135M–0.5B) on a free GPU, complemented by a re-analysis of the published 7–8B results of [6].

2 One framework for pruning, quantization and their combinations

For a linear layer with weights $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ and calibration inputs X, post-training compression solves $\min_{\widehat{W}} \|WX - \widehat{W}X\|_F^2$ under a sparsity or precision constraint. With $H = 2XX^\top$ and $\Delta = W - \widehat{W}$, the objective is $\text{tr}(\Delta H \Delta^\top)$, and we report the normalized layer error $E(\widehat{W}) = \text{tr}(\Delta H \Delta^\top)/\text{tr}(WHW^\top)$. OBS-type methods process the input columns in order. When column j is replaced by its compressed value, the error is propagated to the remaining columns through the inverse Hessian, so later columns compensate for earlier decisions. SparseGPT zeroes, per block of 128 columns, the weights with the smallest saliency $w^2/[H^{-1}]_{jj}^2$; GPTQ rounds each column to a min–max asymmetric grid fitted per row and per group of 128 columns. We implement both as options of one sweep and define three combinations:

- **Joint** (as in [10, 1]): one sweep; in each block the mask is selected first, the grid is fitted on the kept weights, then every column is pruned and quantized before compensation.
- **P→Q**: a pruning sweep returns W_P and its mask M; a quantization sweep on W_P keeps M fixed, so pruned weights stay exactly zero while their error is still compensated.
- **Q→P**: a quantization sweep returns W_Q and its grid G; a pruning sweep on W_Q selects the mask and projects every compensated weight back onto G.

All three have the same storage cost. With sparsity s and bit-width b (16-bit baseline), $\text{TCr} = 1 - \frac{b}{16}(1-s)$ [6]. Joint and P→Q are *mask-first*: the mask is known when the grid is fitted. Q→P is *grid-first*. Layers are compressed sequentially, each calibrated on the outputs of the already-compressed previous layers.

3 Measuring interaction

Layer level. Let E_P, E_Q and E_{PQ} be the layer errors of pruning only, quantization only and a combination. The combination is *additive* if $E_{PQ} = E_P + E_Q$; we define the interaction $I_E = E_{PQ} - (E_P + E_Q)$.

Decomposition. Write $\langle A, B\rangle_H = \text{tr}(AHB^\top)$ and $\|A\|_H^2 = \langle A, A\rangle_H$. Consider a mask M and a fixed grid $q(\cdot)$ without compensation. Let $P = W \odot M$ be the pruned weights and $R = W - q(W)$ the rounding residual, split as $R = R_M + R_{\bar{M}}$ on the pruned and kept entries. Then $\Delta_P = P$, $\Delta_Q = R$ and $\Delta_{PQ} = P + R_{\bar{M}}$, which gives, up to the common normalization,

$$I_E \;\propto\; \underbrace{-\|R_M\|_H^2}_{\text{absorption} \le 0} \;+\; \underbrace{2\langle P, R_{\bar{M}}\rangle_H - 2\langle R_M, R_{\bar{M}}\rangle_H}_{\text{cross terms}}.$$

The first term is never positive: rounding errors on weights that are pruned anyway disappear. Because P, R_M and $R_{\bar{M}}$ have disjoint supports within each row, both cross terms vanish when H is diagonal, i.e. when input features are uncorrelated. In that case $E_{PQ} \leq E_P + E_Q$: *pruning and quantization are sub-additive*. The absorption argument requires that the grid can represent the pruned state; a mask-first order fits the grid to the kept weights, whereas a grid-first order forces the compensated kept weights back onto a grid fitted before pruning, which adds rounding error outside this decomposition. We therefore expect mask-first orders to be sub-additive and the grid-first order to lose this guarantee. With OBS compensation the residuals no longer have these closed forms, so Section 5 tests the prediction.

Model level. With semantic retention Sr, the ratio of compressed to original performance summed over tasks [6], independent degradations multiply. We use $I_{Sr} = Sr_{PQ} - Sr_P\, Sr_Q$; a positive value means the combination does better than independence predicts.

4 Experimental setup

We compress SmolLM2-135M and -360M [11] and Qwen2.5-0.5B [12]: all linear layers of all decoder blocks; embeddings and output head stay in 16 bits. Calibration uses 128 random windows of 2048 tokens from C4 [13], damping $0.01\,\overline{\mathrm{diag}(H)}$, blocks and groups of 128 columns. The grid crosses $s \in \{25, 33.3, 50\}\%$ with $b \in \{4, 3\}$, each in the three orders, plus single-method baselines ($b = 8$ included): 25 models per network. We report WikiText-2 [14] perplexity (2048-token windows) and zero-shot accuracy on ARC-Easy [15], PIQA [16] and HellaSwag [17] (first 500 items each) with lm-eval [18]. On SmolLM2-135M, 18 extra runs repeat the key cells with two more calibration seeds. Everything runs on free T4 GPUs in 16-bit; six SmolLM2-360M runs that overflowed were repeated in 32-bit (three 16-bit controls repeated in 32-bit differ by at most 0.4% in perplexity). Unit checks run at every layer: exact mask ratio, pruned weights still zero after quantization, at most 2^b levels per group, and exact reproduction of the dense forward pass. Code and raw results: `https://github.com/antounyaacoub/esann2027-compose` (DOI 10.5281/zenodo.23396063).

5 Results

Q1: mask-first orders are sub-additive. Fig. 1 shows the relative interaction in all 54 combined models. Both mask-first orders are sub-additive in every one of the 36 cases: by 6–24% for the joint sweep and 3–18% for P→Q. The grid-first order Q→P is super-additive in all nine 4-bit cases (+1 to +7%), as the decomposition predicts once the grid no longer matches the kept weights. At 3 bits Q→P becomes sub-additive for a different reason: 3-bit rounding already sets 31–33% of the weights exactly to zero (15–17% at 4 bits), so a grid-first pruning of 25% only selects weights that are already zero. The resulting model

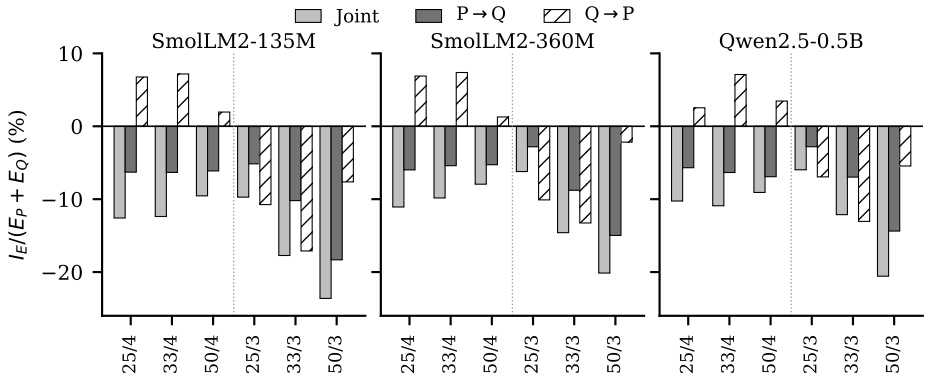


Fig. 1: Layer-level interaction I_E, relative to the additive prediction $E_P + E_Q$ (mean over all linear layers), for each model and each cell (s/b). Negative: sub-additive.

is identical to 3-bit quantization alone ($E_{PQ} = E_Q$, $I_E = -E_P$, same perplexity on all three networks): absorption in its extreme form. The TCr formula, which counts the two compressions independently, misses this free sparsity.

Q2: the order matters at equal compression. Table 1 compares the three orders at identical TCr. Quantizing first is the worst order in 16 of 18 settings, and its penalty grows with compression: at 50%+3-bit it reaches 1.7× (SmolLM2-135M) and 2.0× (SmolLM2-360M) the perplexity of the best order. Among mask-first orders, the joint sweep is best on SmolLM2-360M and Qwen2.5-0.5B (11 of 12 settings) and P→Q on SmolLM2-135M (5 of 6). On SmolLM2-135M at 25%+4-bit, three calibration seeds give 21.75 ± 0.22 for P→Q, 23.35 ± 0.29 for joint and 23.89 ± 0.22 for Q→P: the ranking is well above calibration noise. At 50%+4-bit the joint and P→Q intervals overlap (42.6 ± 2.6 vs. 41.3 ± 1.8), while Q→P stays worse (47.6 ± 2.0). The joint sweep always has the smallest layer error on 4-bit cells, yet on SmolLM2-135M it does not give the smallest perplexity: a smaller local objective does not guarantee a smaller end-to-end loss. Finally, at 3 bits a mask-first combination can beat quantization alone at higher compression: on SmolLM2-360M, 25% pruning + 3-bit gives 20.11 against 25.92 for 3-bit alone. A plausible reading is that the zeros which 3-bit rounding creates anyway are better chosen by OBS saliency than by rounding.

Q3: interaction across scale. The order effect shrinks with model size. At 4 bits, the spread between the best and worst order is 10–24% on SmolLM2-135M, 9–14% on SmolLM2-360M and 2–5% on Qwen2.5-0.5B; the last model also differs in architecture and training, so this trend is suggestive rather than causal. Table 2 reports the model-level interaction of the sequential pipeline. At 25%+4-bit, I_{Sr} stays at or below +0.012 from 135M to 7B, inside the noise of 500-item task subsets ($\approx \pm 0.02$): at moderate compression the two methods compose multiplicatively at every scale we observe. At 33%+3-bit the small

Model	$s + b$	TCr	P only	Q only	Joint	P→Q	Q→P
SmolLM2-135M	25% + 4	0.812	16.48	21.77	23.04	**21.86**	24.03
(dense 15.42)	33% + 4	0.833	18.39	21.77	26.08	**24.11**	27.73
	50% + 4	0.875	32.89	21.77	45.45	**40.15**	49.68
	25% + 3	0.859	16.48	58.17	60.87	**56.88**	58.17
	33% + 3	0.875	18.39	58.17	63.18	**58.97**	68.00
	50% + 3	0.906	32.89	58.17	**88.24**	97.97	150.10
SmolLM2-360M	25% + 4	0.812	12.07	13.29	**13.17**	13.32	14.34
(dense 11.46)	33% + 4	0.833	12.86	13.29	**14.12**	14.20*	15.53
	50% + 4	0.875	18.64	13.29	**20.82**	21.02	23.82
	25% + 3	0.859	12.07	25.92	**20.11***	20.77	25.92
	33% + 3	0.875	12.86	25.92	**20.34***	20.88	30.50*
	50% + 3	0.906	18.64	25.92	**28.71***	30.07*	57.80
Qwen2.5-0.5B	25% + 4	0.812	13.63	14.24	**14.68**	14.71	15.02
(dense 13.07)	33% + 4	0.833	14.52	14.24	**15.62**	15.69	15.99
	50% + 4	0.875	20.21	14.24	**21.61**	22.44	22.74
	25% + 3	0.859	13.63	21.95	21.96	22.11	**21.95**
	33% + 3	0.875	14.52	21.95	**22.00**	22.33	22.96
	50% + 3	0.906	20.21	21.95	**28.41**	29.69	35.61

Table 1: WikiText-2 perplexity at equal TCr (best order in bold). Single-method columns use the same s or b alone. *Re-run in 32-bit after a 16-bit overflow.

Model	25% + 4-bit	33% + 3-bit	Tasks
SmolLM2-135M	+0.012	+0.053	ARC-E, PIQA, HellaSwag
SmolLM2-360M	+0.009	+0.089	ARC-E, PIQA, HellaSwag
Qwen2.5-0.5B	+0.009	+0.012	ARC-E, PIQA, HellaSwag
Mistral-7B-v0.3	+0.009	−0.005	MMLU-Pro, BBH, MATH [6]
LLaMA-3.1-8B	+0.122 (+0.023)	+0.038 (+0.050)	MMLU-Pro, BBH, MATH [6]

Table 2: Model-level interaction $I_{Sr} = Sr_{PQ} - Sr_P Sr_Q$ for prune-then-quantize. For 7–8B, recomputed from the published scores; in brackets, without MATH.

models gain from the combination (I_{Sr} up to +0.09), while Mistral-7B stays at independence. LLaMA-3.1-8B shows a large positive value at 25%+4-bit, but it is driven by the anomalous collapse of 4-bit LLaMA on MATH (0.2%) [6]; without MATH it falls to +0.02.

6 Conclusion

Inside the second-order post-training pipeline, pruning and quantization compose better than independence predicts, provided the mask is chosen before the quantization grid. A simple decomposition explains why: rounding errors on pruned weights are absorbed, and the remaining cross terms vanish for uncorrelated inputs. All 36 mask-first combinations confirm it. Quantizing first breaks this guarantee and is the worst order in 16 of 18 settings, up to twice the best perplexity. In practice: never quantize first; prefer the joint sweep from a few hundred million parameters up. Two further observations deserve follow-up:

low-bit rounding creates a large share of free zeros that TCr ignores, and the smallest layer error does not always give the best model. Extending the analysis to compensated sweeps, semi-structured sparsity and larger models is left for future work.

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